

Basket of Electives

Programme	B. Sc. Computer Science				
Course Title	Mathematical and Statistical Foundations for Data Science				
Type of Course	Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basic Mathematics and Statistics 2. Python basics (If Python implementation is preferred in module V by the course tutor)				
Course Summary	This undergraduate course provides the fundamental mathematical and statistical tools necessary for understanding and analyzing data in the context of data science. The course covers topics ranging from basic algebraic operations to advanced statistical techniques.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply Vector and Matrix operations to solve computational problems.	Ap	P	Instructor-created exams / Assignment

CO2	Students will evaluate eigenvalues and eigenvectors to decompose matrices, enabling them to analyze and interpret data transformations effectively.	An	P	Instructor-created exams / Assignment
CO3	Students will apply fundamental probability concepts to solve real-world problems.	Ap	P	Assignment / Quiz
CO4	Students will utilize statistical techniques for data interpretation and decision-making	Ap	P	Instructor-created exams / Assignment
CO5	Students will apply sampling techniques and hypothesis tests to make inferences about populations from sample data, using one-tailed, two-tailed tests, and ANOVA for analysis	Ap	Q	Assignment / Case Studies
CO6	Students will apply PCA to reduce data dimensionality, identify principal components, and interpret results in data science application.	Ap	R	Assignment / Case Studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks 70
I	Linear Algebra		14	20

	1	Scalars, Vectors, Matrices and Tensors Vectors:- Vector Arithmetic - Vector Addition, Vector Subtraction, Vector Multiplication, Vector Division; Vector Dot Product; Vector Scalar Multiplication	2	
	2	Matrix Multiplication, Identity and Inverse Matrices, Linear dependence and Span , Norms, Diagonal and Orthogonal Matrices	4	
	3	Eigenvectors and Eigenvalues , Eigen Decomposition	2	
	4	Singular Value Decomposition	2	
	5	The Trace Operator , The Determinant	2	
	6	Principal Component Analysis	2	
II	Probability		14	20
	7	Random Variables , Probability Distributions	3	
	8	Marginal Probability , Conditional Probability,	1	
	9	The Chain Rule of Conditional Probabilities	3	
	10	Independence and Conditional Independence	2	
	11	Expectation, Variance and Covariance	1	
	12	Common Probability Distributions - Bernoulli Distribution	3	
		Binomial, Normal and Poisson Distribution		
	13	Bayes' Rule	1	
III	Basic Statistics		8	15
	14	Measures of Central Tendency	3	
	15	Measures of Dispersion	2	

	16	Skewness, Kurtosis	1	
	17	Correlation and Regression	2	
IV	Sampling and Hypothesis Testing		12	15
	18	Sampling distributions of the sample mean and the sample variance for a normal population	2	
	19	Point and interval estimation	1	
	20	Sampling distributions (Chi-square, t, F, Z)	3	
	21	Hypothesis testing	1	
	22	One tailed and two-tailed tests, Analysis of variance, ANOVA, One way and two way classifications.	5	
V	Open ended module		12	
	Solve the following problems mathematically. Or Try to implement these problems using Python.			
	1	Linear Algebra Concepts to be learned:- • Vector arithmetic (a) Define a vector a and b with the length of 3 and the integer values 1, 2 and 3. (b) Perform addition, subtraction, multiplication, division and dot product of the two vectors a and	1	

	2	Linear Algebra Concepts to be learned: - Matrix arithmetic (a) Create two 2 row, 3 column matrices, say A and B . Perform matrix addition, subtraction, division and multiplication (element-wise matrix multiplication or the Hadamard product). (b) Create a matrix A with 3 rows and 2 columns, and a matrix B with 2 rows and 2 columns. Perform matrix dot product of matrices A and B .	1																			
	3	Linear Algebra Concepts to be learned: - - Singular Value Decomposition - Orthogonal Matrices - Diagonal Matrix - Singular Value - Eigen values and Eigen Vectors - Matrix Multiplication Find the singular value decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$	2																			
	4	Basic Statistics Concepts to be learned: - Measures of Central Tendency - Mean, Median, ModeFind the mean, the median, and the mode for the number of vehicles owned in a s survey of 52 households. <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>f</td><td>2</td><td>12</td><td>15</td><td>11</td><td>6</td><td>3</td><td>1</td><td>2</td></tr></table>	x	0	1	2	3	4	5	6	7	f	2	12	15	11	6	3	1	2	1	
x	0	1	2	3	4	5	6	7														
f	2	12	15	11	6	3	1	2														

	5	Basic Statistics Concepts to be learned: - Measures of Dispersion - Range, Variance, Standard Deviation Find the range, the variance and the standard deviation for the sample of ten IQ scores randomly selected from a school for academically gifted students. 142 152 138 145 148 139 147 155 150 153	1												
	6	Application of Probability Concepts to be learned: - - Probability basics - Combinations - Mutually exclusive events - Complementary events Of 10 girls in a class, 3 have blue eyes. If two of the girls are chosen at random, what is the probability that (i) both have blue eyes, (ii) neither has blue eyes, (iii) at least one has blue eyes?	1												
	7	Application of Probability Concepts to be learned: - - Probability Basics - Contingency Tables - Marginal and Joint Probabilities - Conditional Probability The following two-way contingency table gives the breakdown of the population in a particular locale according to age and tobacco usage. <table><tr><th rowspan="2">Age</th><th colspan="2">Tobacco Use</th></tr><tr><th>Smoker</th><th>Non-smoker</th></tr><tr><td>Under 30</td><td>0.05</td><td>0.20</td></tr><tr><td>Over 30</td><td>0.20</td><td>0.55</td></tr></table>	Age	Tobacco Use		Smoker	Non-smoker	Under 30	0.05	0.20	Over 30	0.20	0.55	1	
Age	Tobacco Use														
	Smoker	Non-smoker													
Under 30	0.05	0.20													
Over 30	0.20	0.55													

		A person is selected at random. Find the probability of each of the following events. (a) The person is a smoker. (b) The person is under 30. (c) The person is a smoker who is under 30.		
	8	Application of Probability Concepts to be learned: - - Understand the characteristics of a normal distribution. - Calculating and interpreting z-scores. Suppose the heights H of 800 students are normally distributed with mean 66 inches and standard deviation 5 inches. Find the number N of students with heights (a) between 65 and 70 inches, (b) greater than or equal to 6 feet(72inches).	1	
	9	Application of Probability Concepts to be learned: - Bayes' Theorem A patient goes to see a doctor. The doctor performs a test with 99 percent reliability--that is, 99 percent of people who are sick test positive and 99 percent of the healthy people test negative. The doctor knows that only 1 percent of the people in the country are sick. If the patient tests positive, what are the chances the patient is sick?	1	
	10	Sampling and Hypothesis Testing Concepts to be learned: - - Hypothesis testing - Contingency tables, and - Chi-square analysis	1	

		A die is suspected of being biased. It is rolled 25 times with the following result: <table><thead><tr><th><u>Outcome</u></th><th><u>Frequency</u></th></tr></thead><tbody><tr><td>1</td><td>9</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>1</td></tr><tr><td>4</td><td>8</td></tr><tr><td>5</td><td>3</td></tr><tr><td>6</td><td>0</td></tr></tbody></table> Conduct a significance test to see if the die is biased. (a) What Chi Square value do you get and how many degrees of freedom does it have? (b) What is the p value?	<u>Outcome</u>	<u>Frequency</u>	1	9	2	4	3	1	4	8	5	3	6	0		
<u>Outcome</u>	<u>Frequency</u>																	
1	9																	
2	4																	
3	1																	
4	8																	
5	3																	
6	0																	
	11	Sampling and Hypothesis Testing Concepts to be learned: - • Central Limit Theorem • Sampling distribution of the Sample Mean • Standard Error of the Mean • Z-score • Normal Distribution Properties • Probability Calculation Suppose scores on an IQ test are normally distributed, with a mean of 100. Suppose 20 people are randomly selected and tested. The standard deviation in the sample group is 15. What is the probability that the average test score in the sample group will be at most 110?	1															

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6
CO 1	3	-	2	2	2	2
CO 2	3	-	2	3	2	2
CO 3	3	-	3	3	2	2
CO 4	3	-	3	3	2	2
CO 5	3	-	3	3	2	2
CO 6	3	-	3	3	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4	✓	✓		✓
CO 5	✓	✓	✓	✓
CO 6	c✓	✓	✓	

References

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press, 2017.
2. Gilbert Strang. Introduction to Linear Algebra. 5th ed. Wellesley-Cambridge Press, 2016.
3. S. Ross, Introduction to Probability and Statistics for and Engineers and Scientists, Third Edition, Elsevier, 2004.